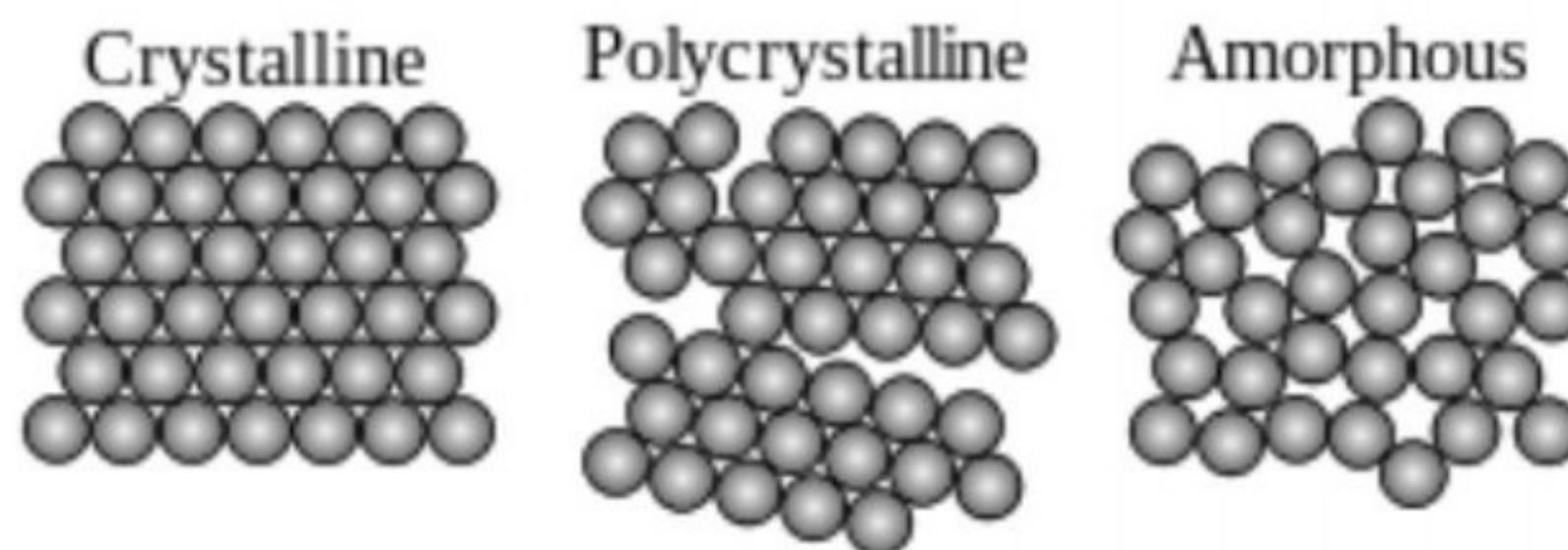


Solid State Physics

Crystalline and non-crystalline solid:

In crystalline solids, the atoms are stacked in a regular manner in crystals. In this type of crystals, the atomic array is periodic, that is, a representative unit or motif is repeated at regular intervals along any and all directions in the crystal.

On the other hand, in non-crystals, the same atomic groups are arrayed more randomly.



Single crystal and polycrystalline solids:

In single crystals, the periodicity extends throughout the material.

However, in polycrystalline crystals, the periodicity does not extend throughout the crystal but is interrupted at grain boundaries.

Lattice:

The regular periodic arrangements of atoms, ions and molecules in space which looks like a net is called lattice.

Lattice point:

If each atom in a lattice is replaced by a point, then such point is called lattice point, i.e., every point in a regular periodic array in space is called lattice point.

Basis:

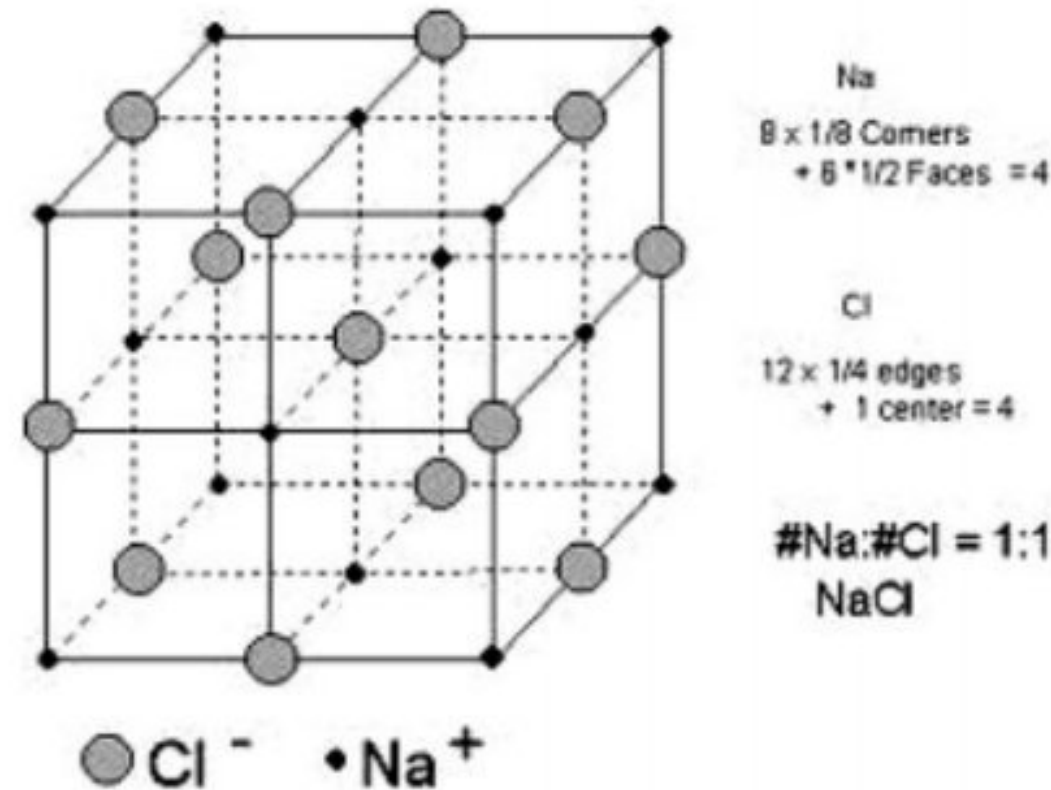
The structure of all crystal can be described in terms of a lattice, with a group of atoms attached to every lattice point. The group of atoms is called basis; it is repeated in space to form the crystal structure.

In other words, a basis is the assimilation of atoms, group of atoms or molecules in ionic crystal identical in composition, arrangement and orientation, it is composed of ions.

Crystal:

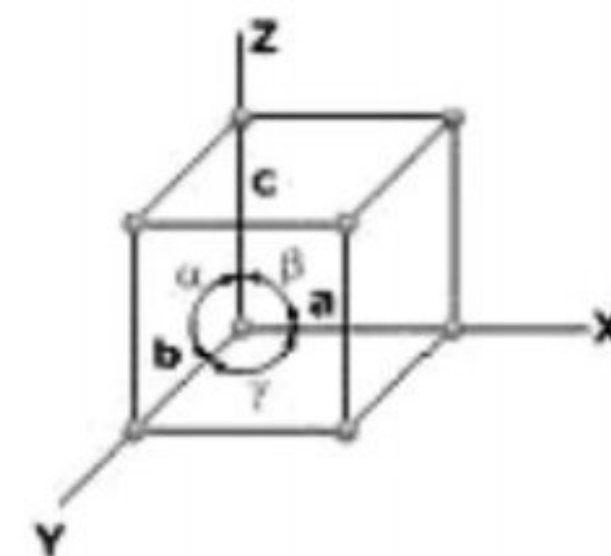
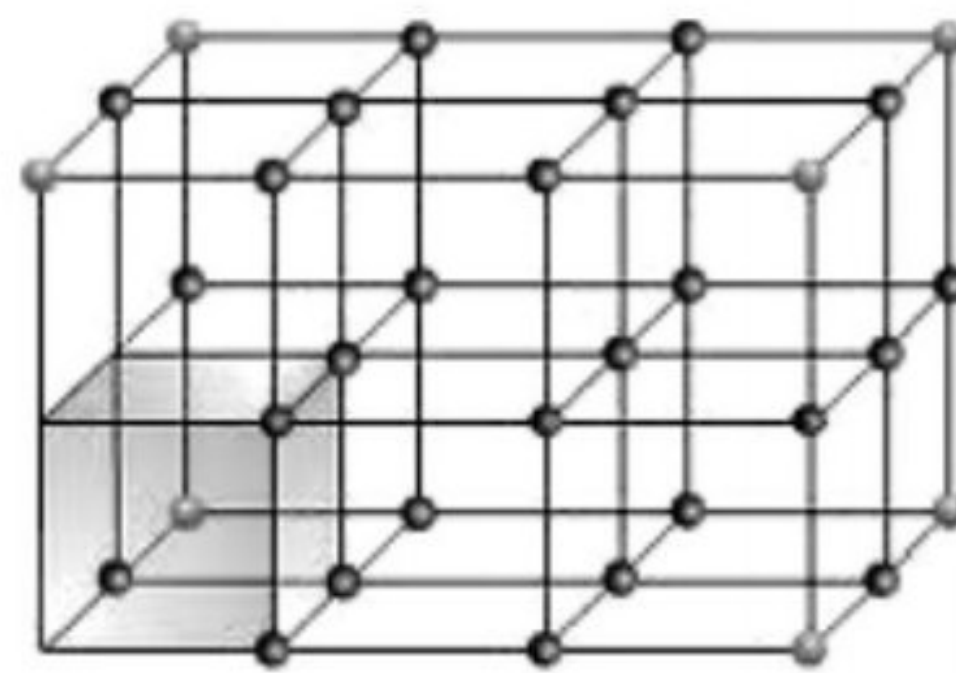
A crystal is a regular array of three dimensional units. The structural units are atoms, ions and molecules or group of atoms.

An ideal crystal is constructed by the infinite repetition of identical structural unit in space. In the simplest crystal the structural unit is a single atom, as in copper, silver, gold, iron, aluminium.



Unit cell:

The lattice array of a group of atoms is a periodic repetition of a fundamental grouping. Therefore it is not necessary to consider the entire array. This is done by selecting a region of the crystal defined by three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ translated by any integral multiple of these vectors, reproduce a similar region of the crystal. This region is called a unit cell and these vectors as basis vectors.



Crystal systems:

On the basis of the lengths and directions of the axes of the symmetry, all the crystal may be classified into the following seven systems:

System	No. of lattices	Lattice symbols	Bravais lattice	Characteristic Symmetry Element	Unit cell characteristic	Examples
Cubic	3	P sc	Simple	Four 3-fold rotation axes (cube diagonals)	$a=b=c$ $\alpha=\beta=\gamma=90^\circ$	NaCl, Cu, Ag
		I bcc	Body centered			
		F fcc	Face centered			
Triclinic	1	P	Simple	None	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma \neq 90^\circ$	CaSO ₄
Monoclinic	2	P	Simple	One 2-fold rotation axes	$a \neq b \neq c$ $\alpha=\beta=90^\circ \neq \gamma$	FeSO ₄
		C	Base centered			
Orthorhombic	4	P	Simple	Three mutually perpendicular 2-fold rotation inversion axes	$a \neq b \neq c$ $\alpha=\beta=\gamma=90^\circ$	MgSO ₄
		C	Base centered			
		I	Body centered			
		F	Face centered			
Tetragonal	2	P	Simple	One 4-fold rotation axes or a 4-fold rotation inversion axes	$a=b \neq c$ $\alpha=\beta=\gamma=90^\circ$	NiSO ₄
		I	Body centered			
Hexagonal	1	P	Simple	One 6-fold rotation axes	$a=b \neq c$ $\alpha=\beta=\gamma \neq 90^\circ$	SiO ₂
Rhombohedral (Trigonal)	1	R	Simple	One 3-fold rotation axes	$a=b=c$ $\alpha=\beta=\gamma \neq 90^\circ$	CuSO ₄

Coordination number:

Every atom in a crystalline structure is surrounded by other atoms. The number of nearest neighbours that surround each atom is called Coordination number.

NaCl structure:

The NaCl crystal has an fcc structure with Na^+ and Cl^- ion as the basis. This is an ionic crystal. Fig.-01 represents a unit cell of NaCl lattice. The Na^+ ions are situated at the corners as well as at the centers of the faces of the cube. The Cl^- ions are relatively displaced half the edges of the unit cell along each axis. Thus NaCl crystal can be thought of as composed of superposed fcc Na^+ and Cl^- sub-lattices.

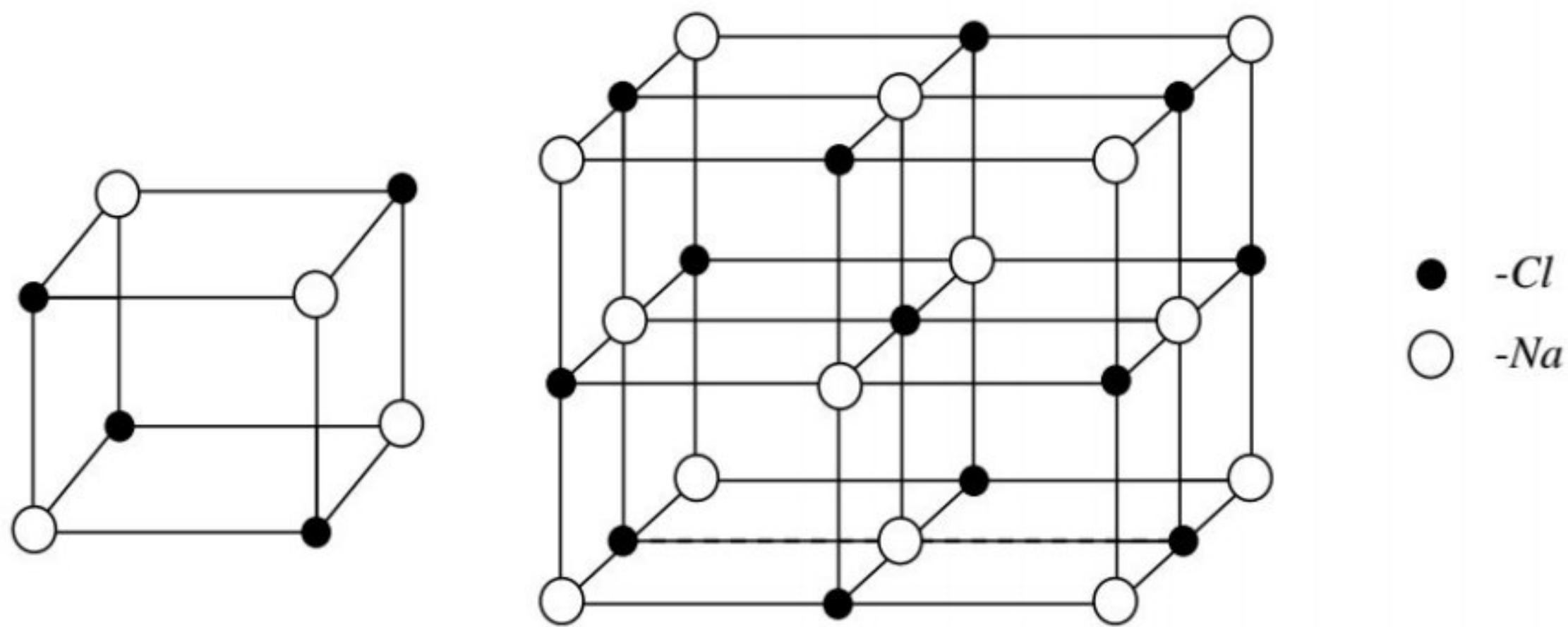


Fig.-01

A cell has 8 corners and 6 faces. So it has $\left(\frac{1}{8} \times 8\right) + \left(\frac{1}{2} \times 6\right) = 4$ ions of one kind and similarly 4 ions of the other kind. Thus there are 4 $\text{Na}^+ - \text{Cl}^-$ ion pair per unit cell. There are four molecules of NaCl in each cube, with ions in the positions:

$$\begin{aligned} \text{Na: } & 000, \frac{1}{2}\frac{1}{2}0, \frac{1}{2}0\frac{1}{2}, 0\frac{1}{2}\frac{1}{2} \\ \text{Cl: } & \frac{1}{2}\frac{1}{2}\frac{1}{2}, 00\frac{1}{2}, 0\frac{1}{2}0, \frac{1}{2}00 \end{aligned}$$

Each Na^+ ion has 6 Cl^- ions as nearest neighbours. Similarly each Cl^- ion has 6 Na^+ ions as nearest neighbours. Hence the coordination number of NaCl for opposite kind of ions is 6 and the nearest neighbour distance is $\frac{a}{2}$.

CsCl structure:

The structure of CsCl is shown in fig.-2. The CsCl crystal has simple cubic structure with Cs^+ and Cl^- ions as the basis. The Cs^+ ions are situated at the corners of the cubic cell and Cl^- ion at the body center of the cubic cell and vice-versa. Thus CsCl crystal may be considered to be a combination of two simple cubic sublattices, one of Cs^+ ion and the other of Cl^- ion.

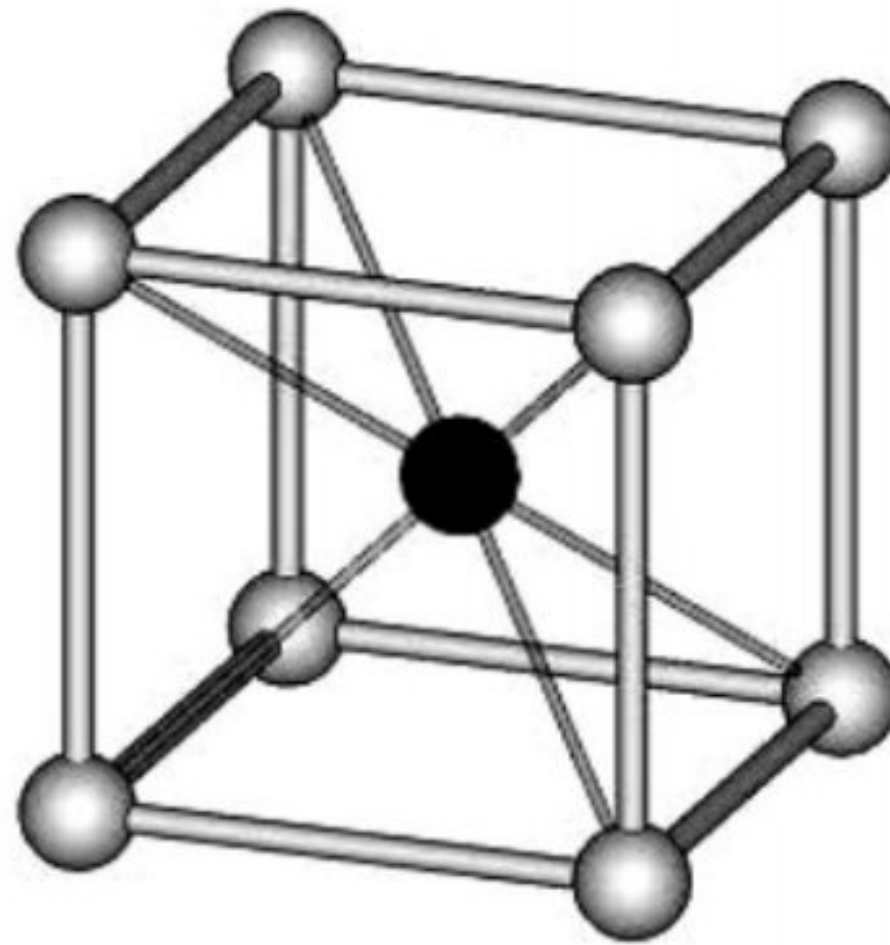


Fig.-02

There is one molecule per primitive cell, with ions at the corners $0,0,0$ and body centered positions $\frac{1}{2}\frac{1}{2}\frac{1}{2}$ of the simple cubic space lattice. The coordination number for same kind of ions is 6 and the nearest neighbour distance is a . The coordination number for opposite kind of ions is 8 and the nearest neighbour distance is $\frac{\sqrt{3}}{4}a$.

Packing fraction:

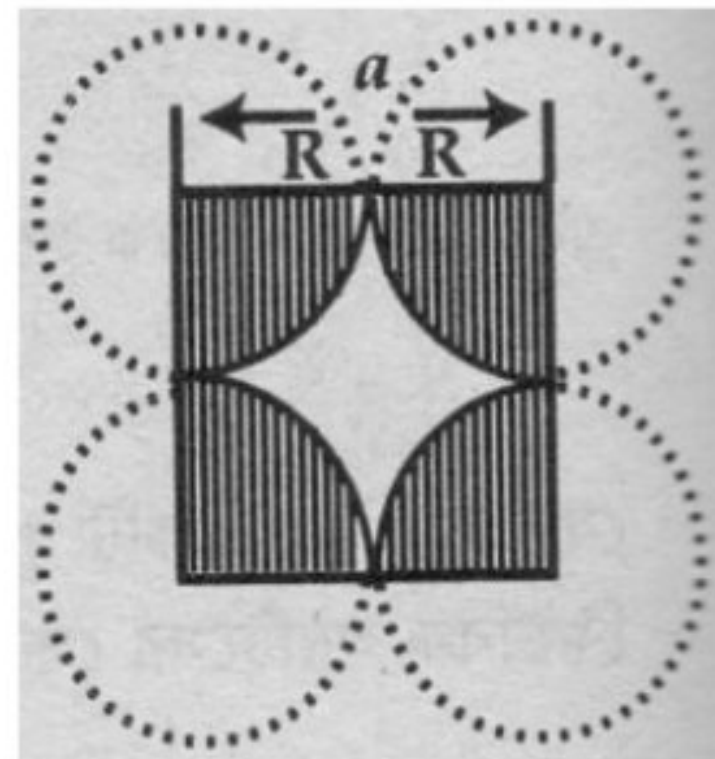
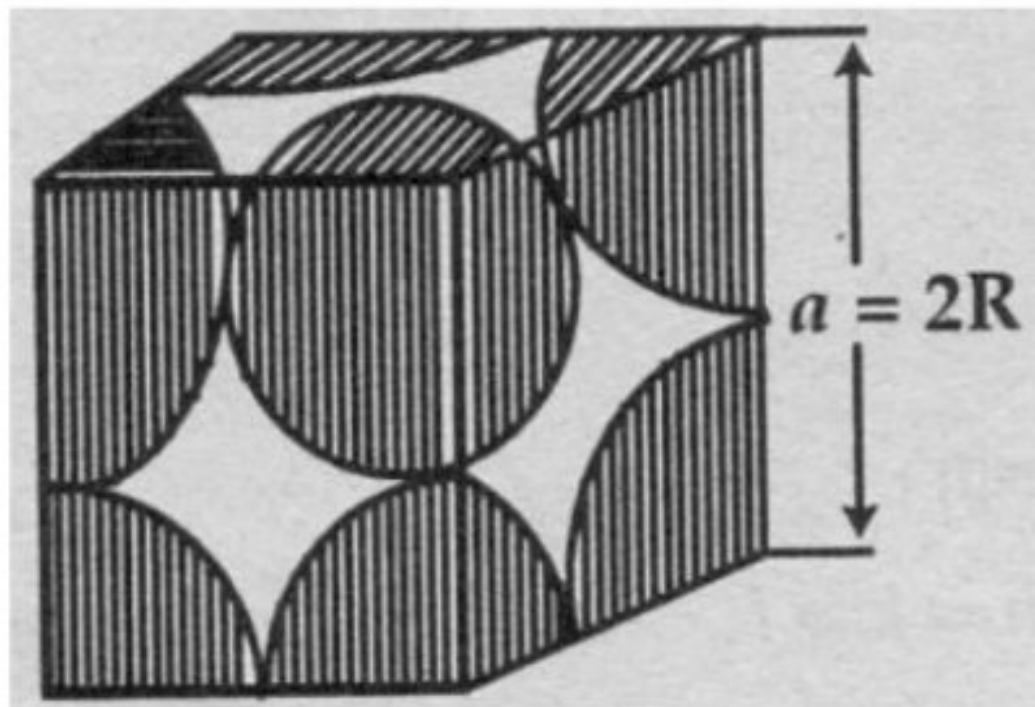
Packing fraction is defined as the ratio of the volume of the atoms per unit cell to the total volume of the unit cell.

$$P.F. = \frac{\text{The volume of the atoms per unit cell}}{\text{The total volume of the unit cell}}.$$

(i) Simple cubic:

There are eight atoms per unit cell situated at the eight corners. If a is the side of the unit cell and r is the radius of the atom, then from figure, $a = 2r$.

[N.B.-R replace to r]



Then, total volume of the unit cell is,

$$= a^3 = (2r)^3 = 8r^3$$

The number of atoms per unit cell

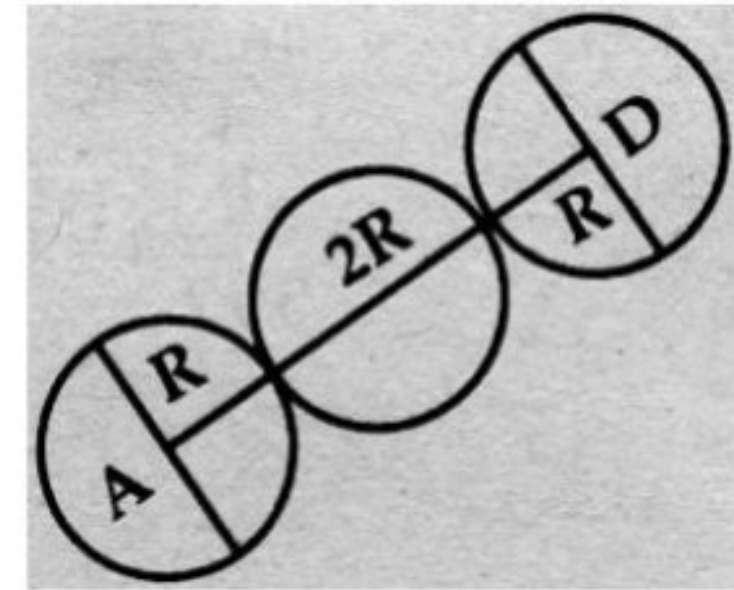
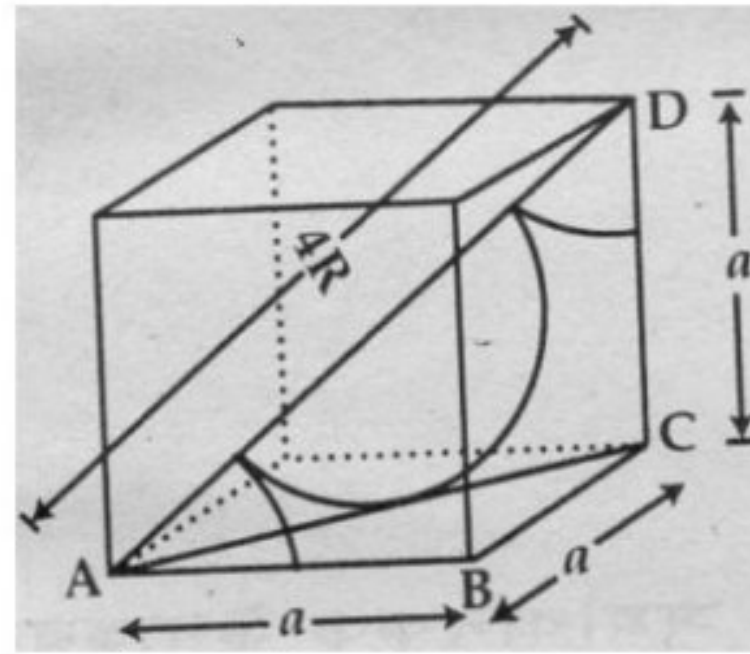
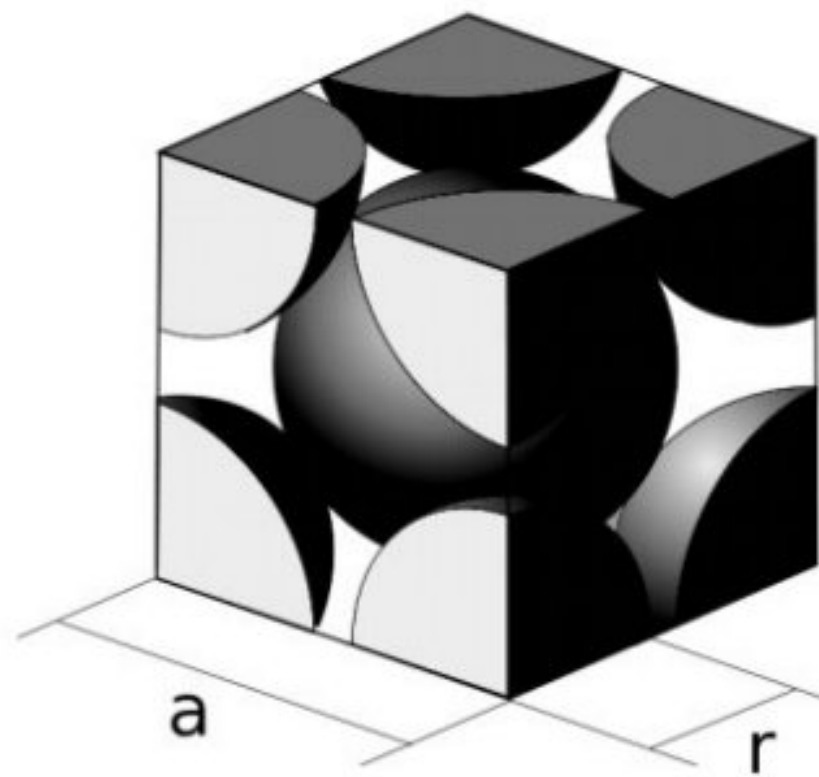
$$= \frac{1}{8} \times 8 = 1$$

$\therefore$ The volume of the atoms per unit cell $= \frac{4}{3} \times \pi r^3$

$$\begin{aligned} \therefore \text{Packing Fraction} &= \frac{\text{The volume of the atoms per unit cell}}{\text{The total volume of the unit cell}} \\ &= \frac{\frac{4}{3}\pi r^3}{8r^3} = \frac{\pi}{6} = 0.53 \end{aligned}$$

(ii) bcc:

In the unit cell of *bcc* lattice, there are eight atoms per unit cell situated at the eight corners and one atom per cell at the body centre.



[N.B.-R replace to r]

Then from figure, it can be written as

$$AC^2 = a^2 + a^2$$

$$AC = a\sqrt{2}$$

$$AD = r + 2r + r = 4r$$

Again, $AD^2 = AC^2 + CD^2$

$$(4r)^2 = a^2 \times 2 + a^2 = 3a^2$$

$$\therefore 16r^2 = 3a^2$$

$$a^2 = \frac{16}{3}r^2;$$

$$a = \frac{4}{\sqrt{3}}r$$

Then, total volume of the unit cell is,

$$= a^3 = \left(\frac{4}{\sqrt{3}}r\right)^3 = \frac{64}{3\sqrt{3}}r^3$$

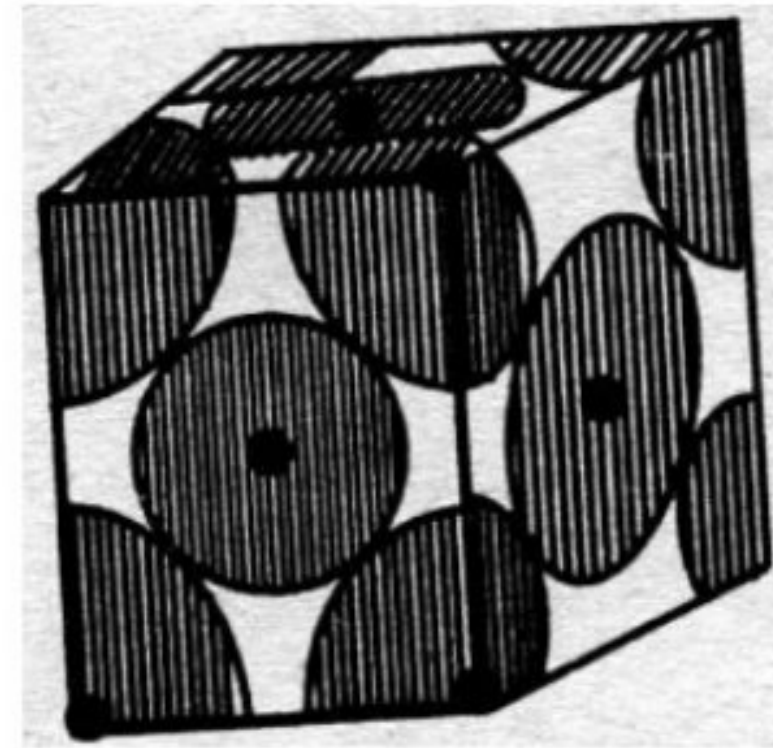
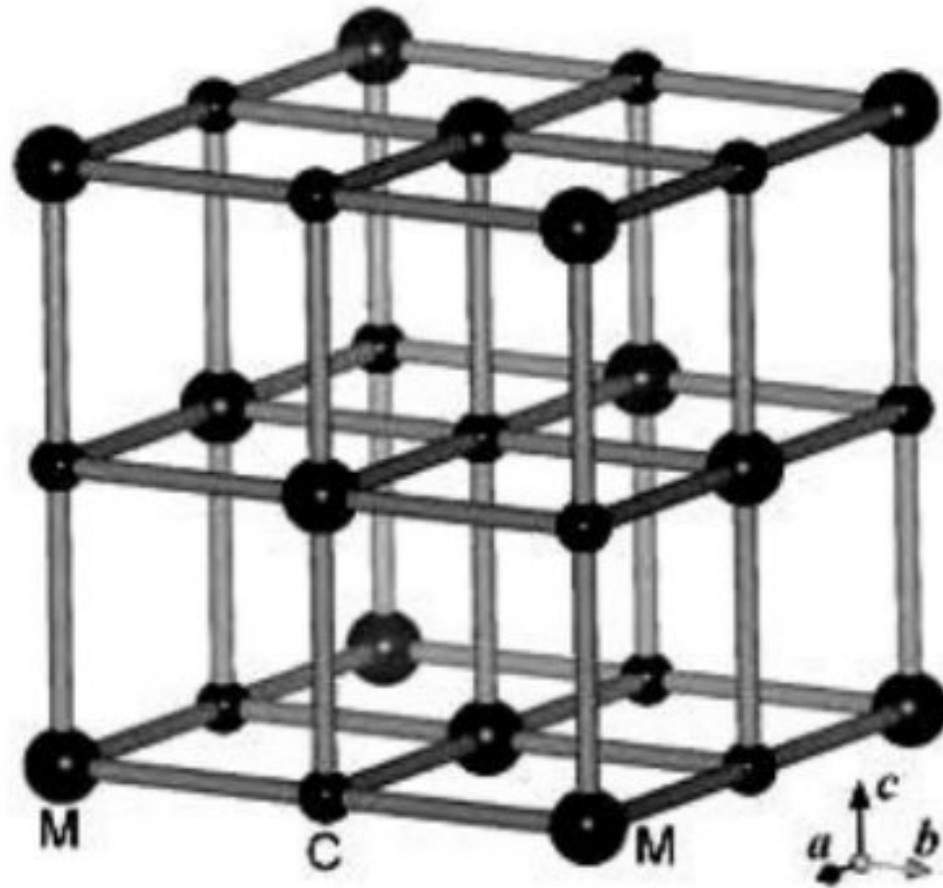
The number of atoms per unit cell = 2

$$\therefore \text{The volume of the atoms per unit cell} = 2 \times \frac{4}{3} \pi r^3 = \frac{8}{3} \pi r^3$$

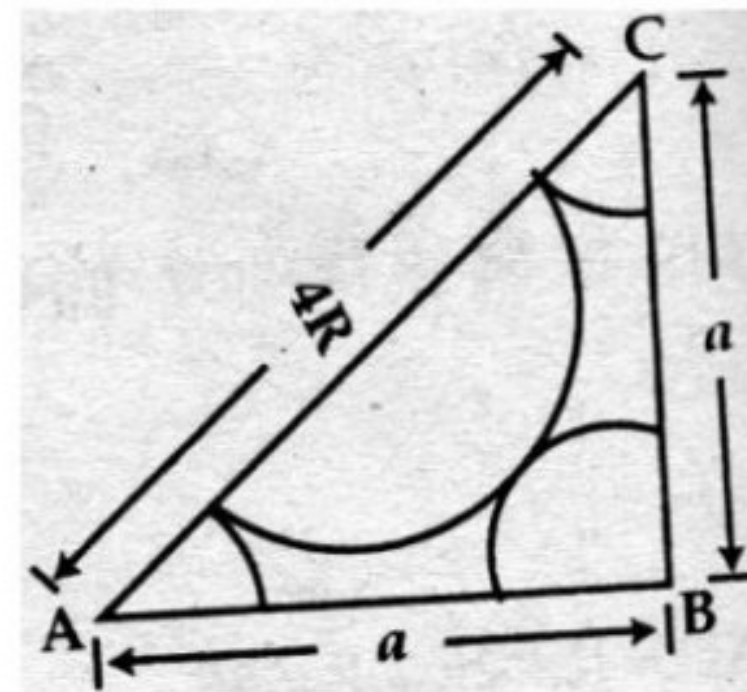
$$\therefore \text{Packing Fraction} = \frac{\frac{8}{3} \pi r^3}{\frac{64}{3\sqrt{3}} r^3} = \frac{\pi\sqrt{3}}{8} = 0.68$$

(iii) *fcc*:

In the unit cell of *fcc* lattice, there are eight atoms per unit cell situated at the eight corners and one atom at the centre of each face.



[N.B.-R replace to r]



Then from figure, it can be written as

$$AC = r + 2r + r = 4r$$

$$AC^2 = a^2 + a^2$$

$$(4r)^2 = 2a^2$$

$$\therefore a^2 = \frac{16r^2}{2}$$

$$a^2 = 8r^2$$

$$a = 2\sqrt{2}r$$

Then, total volume of the unit cell is,

$$= a^3 = (2\sqrt{2}r)^3$$

$$= 16\sqrt{2}r^3$$

The number of atoms per unit cell = 4

$$\therefore \text{The volume of the atoms per unit cell} = 4 \times \frac{4}{3} \pi r^3 = \frac{16}{3} \pi r^3$$

$$\therefore \text{Packing Fraction} = \frac{\frac{16}{3} \pi r^3}{\frac{16\sqrt{2}}{1} r^3} = \frac{1}{3\sqrt{2}} \pi = 0.74$$

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Miller indices:

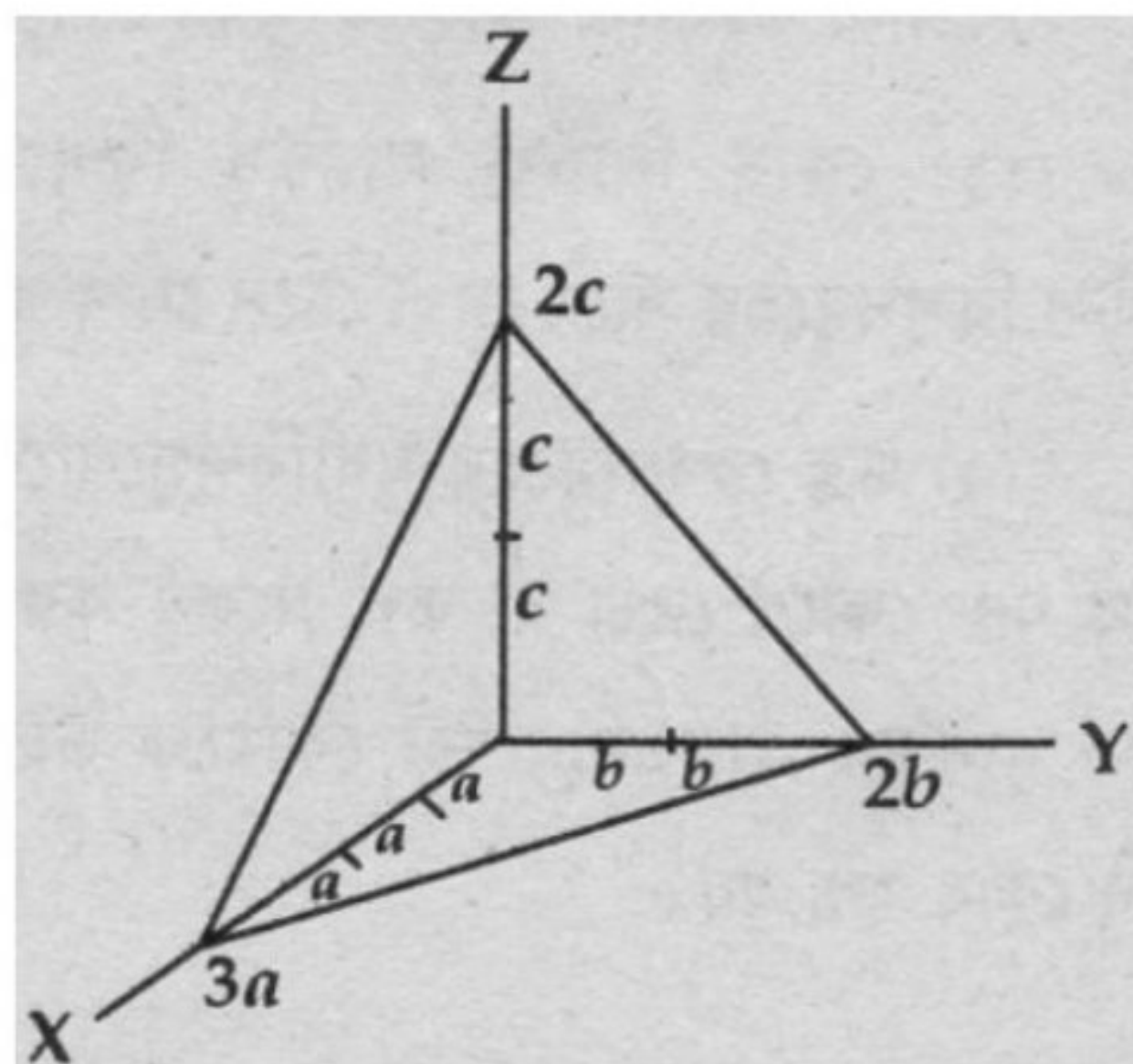
In order to represent any particular crystal plane in terms of the crystallographic axes and the intercepts of the plane on these axes, the reciprocal of the intercepts are taken on the three axes in terms of the lattice constants and these reciprocals are reduced to the smallest three integers having the same ratio. These integers are the Miller indices of the plane.

Procedure for finding Miller indices:

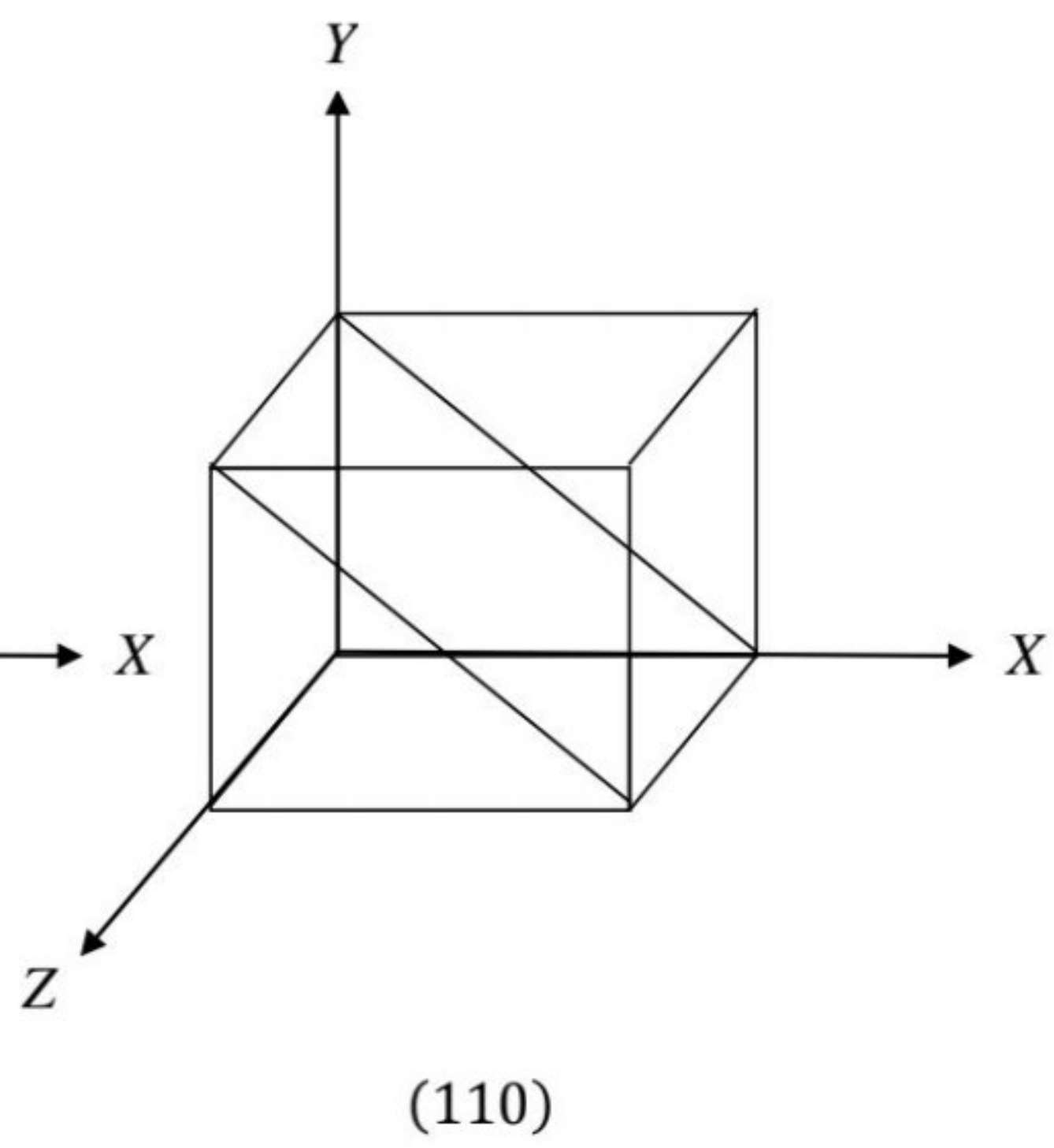
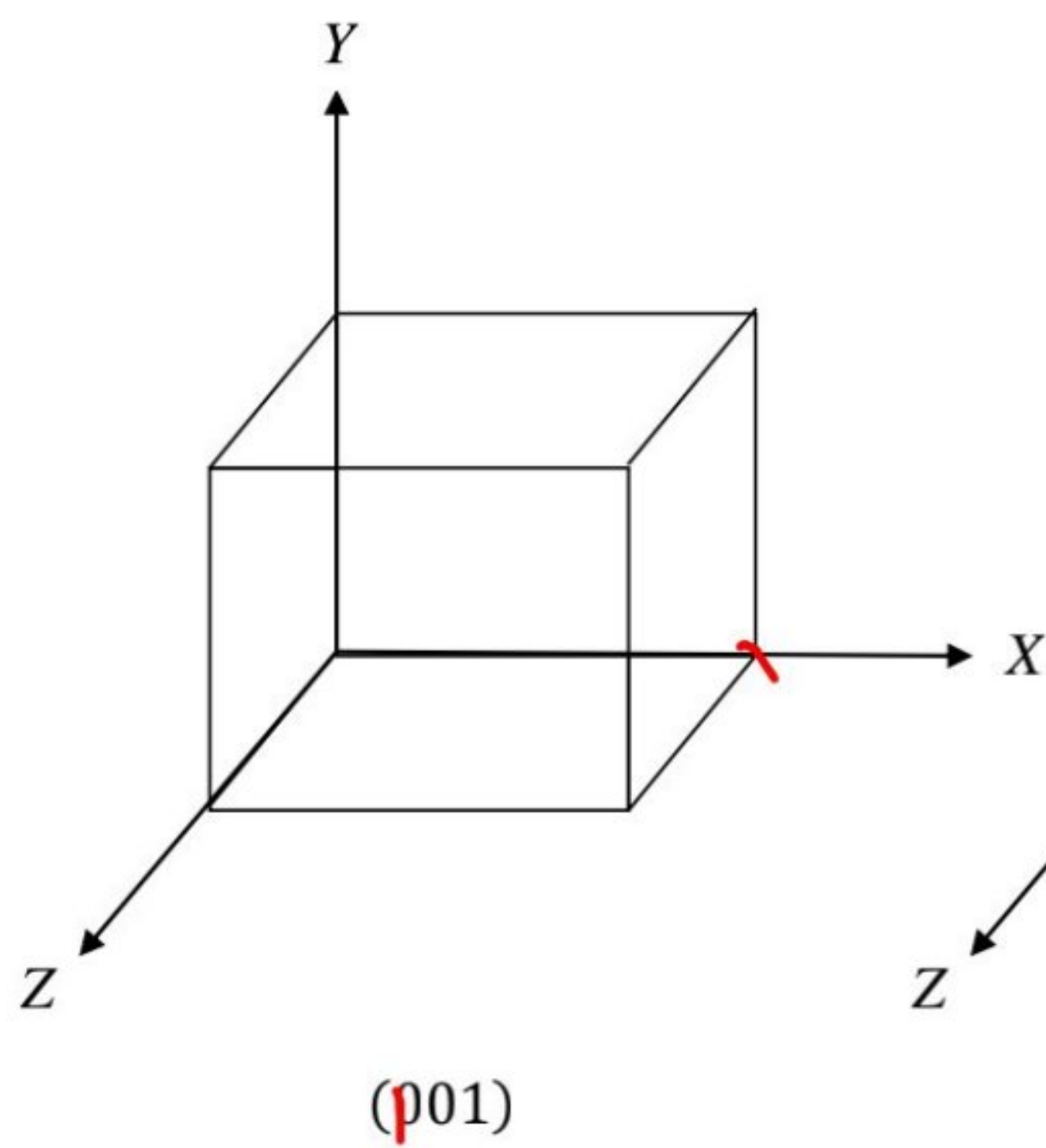
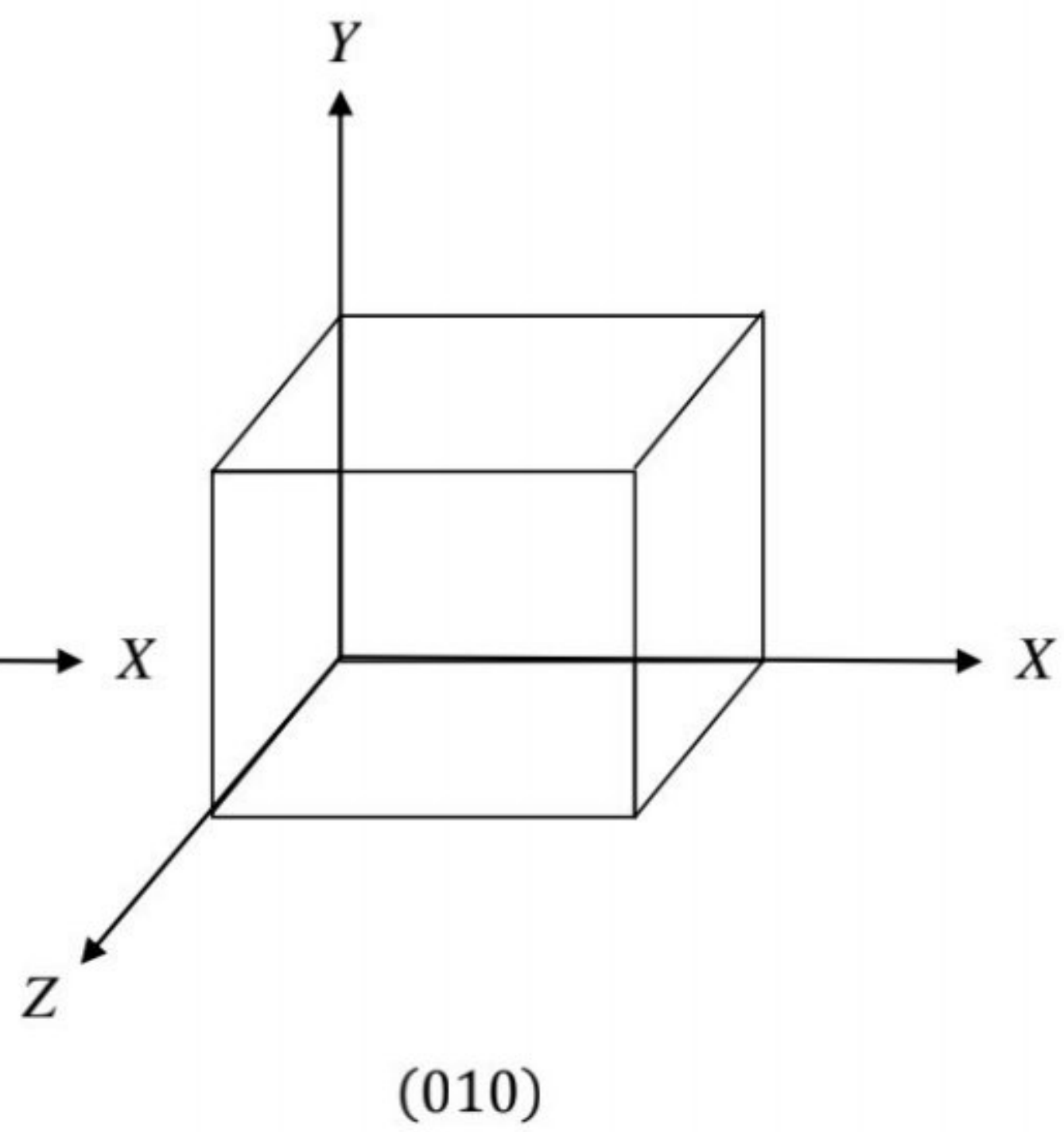
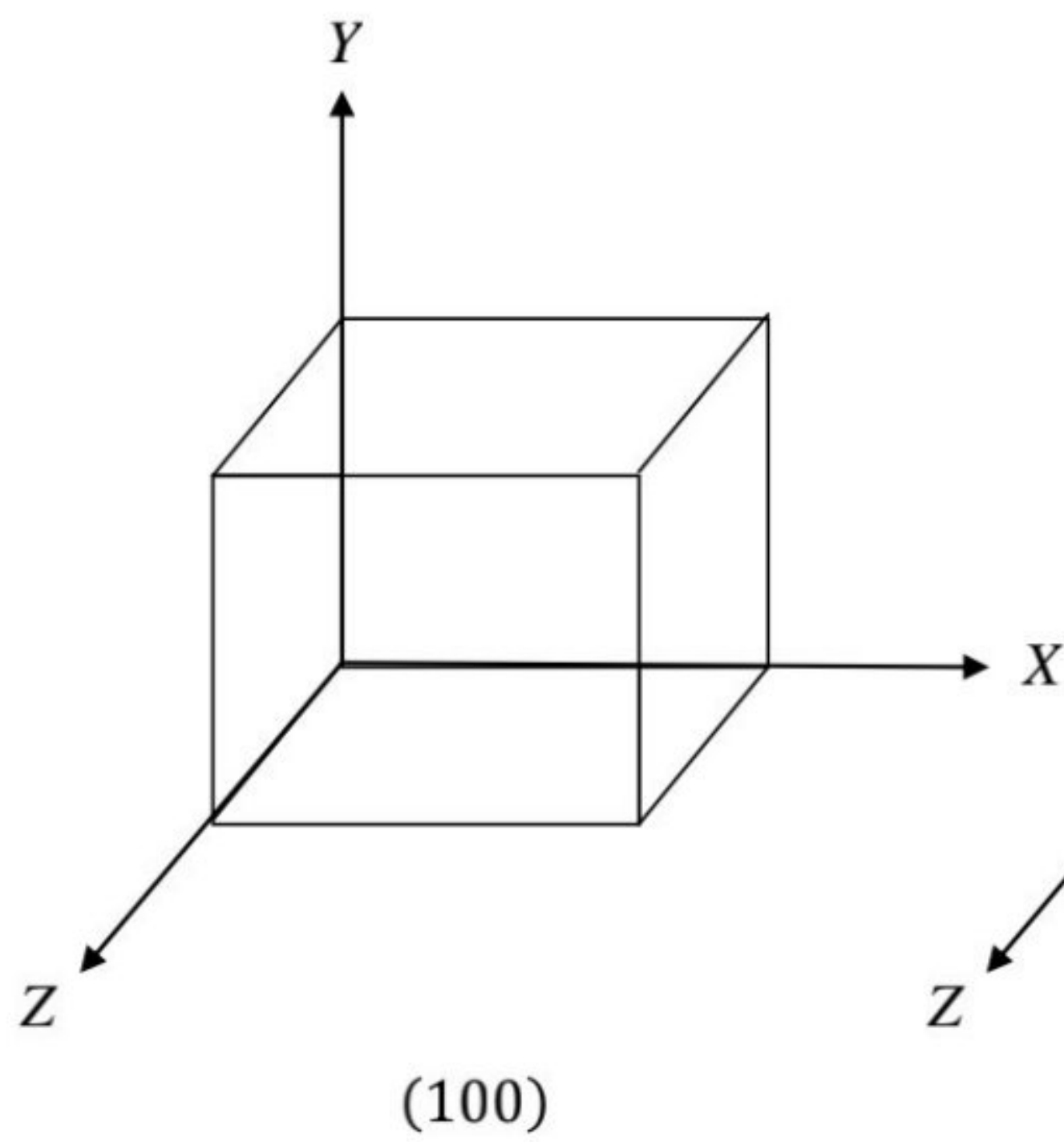
- Determine the intercepts of the plane along the axes X, Y and Z in terms of the lattice constants a , b and c .
- Determine the reciprocals of these numbers.
- Find the least common denominator and multiply each by this. The result is written in the form (hkl) and is called the Miller indices of the plane.

Example:

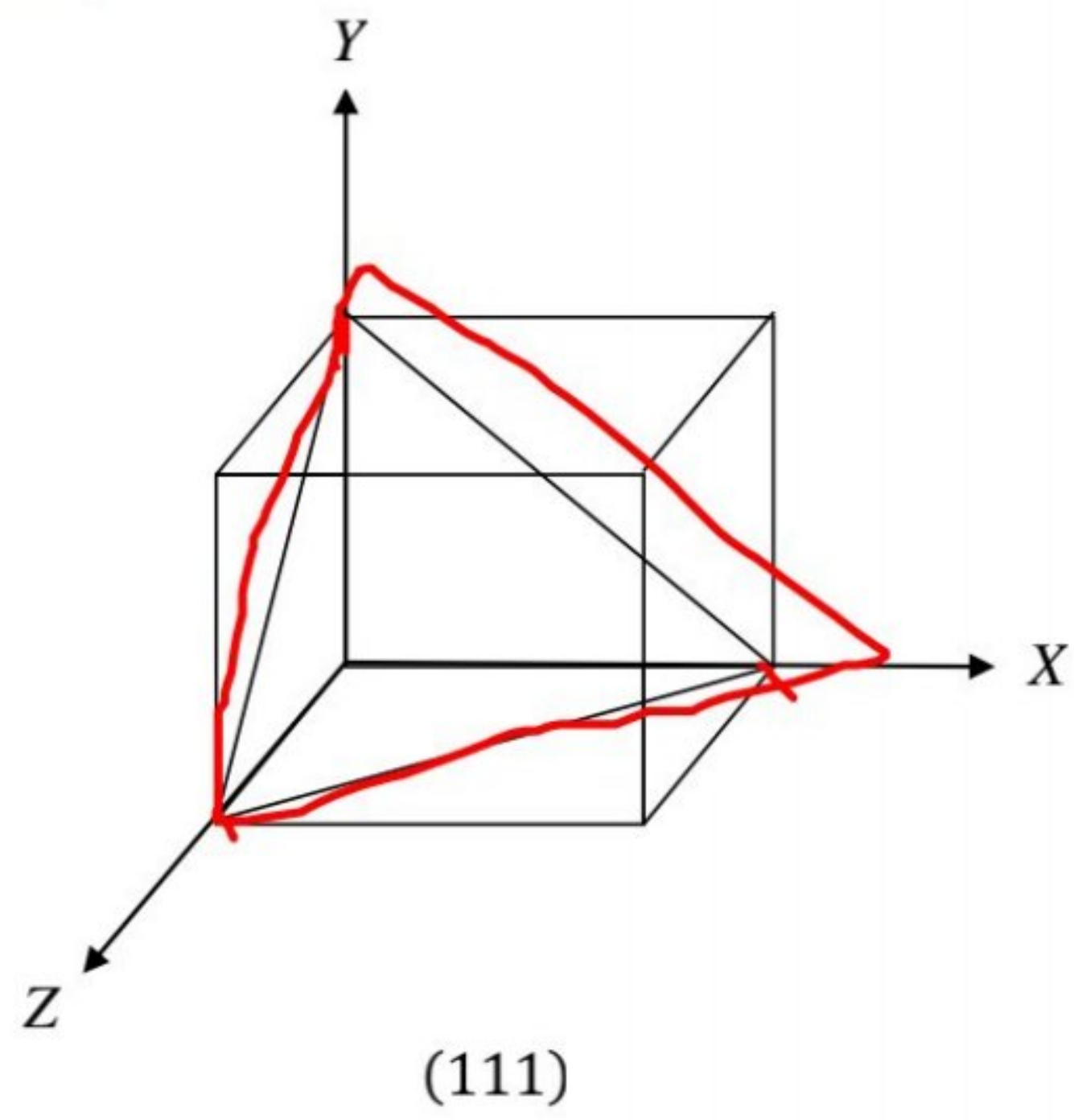
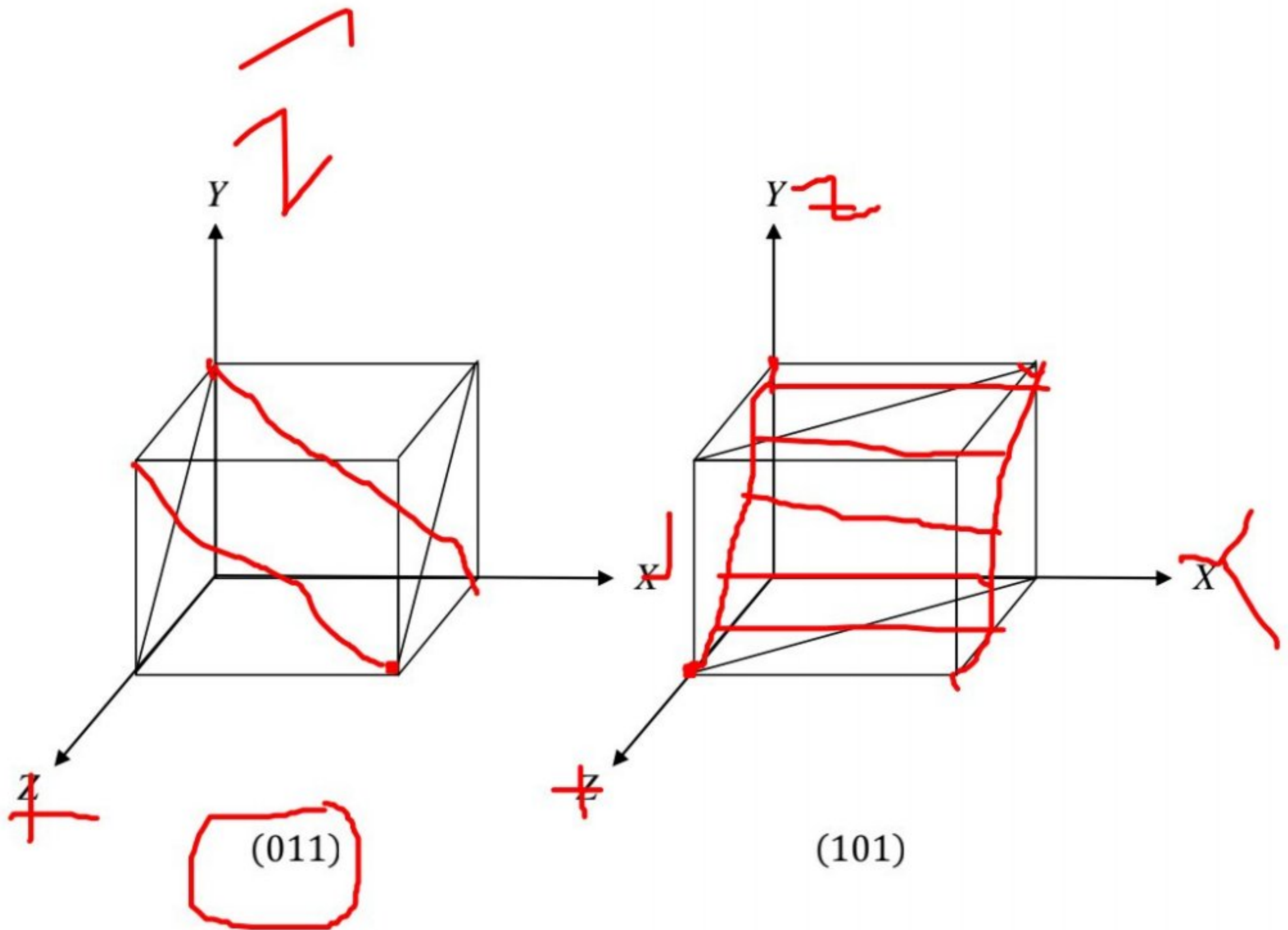
Let us consider a plane which intercepts X, Y and Z axes are $3a$, $2b$ and $2c$. The reciprocals are these values $\frac{1}{3}$, $\frac{1}{2}$ and $\frac{1}{2}$. Multiplying each by 3, we get 2, 3, 3. Hence (233) are the Miller indices of the plane.



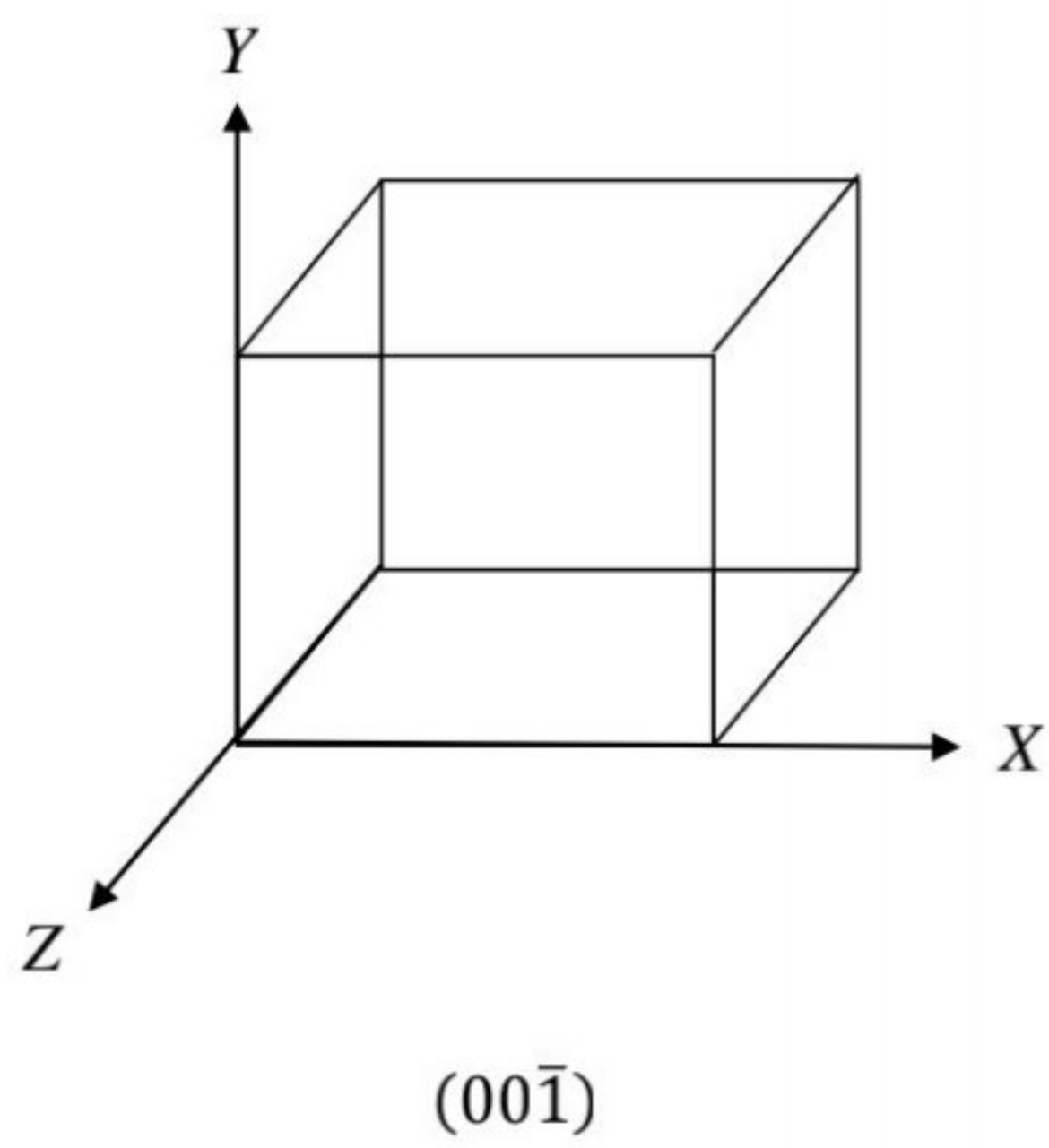
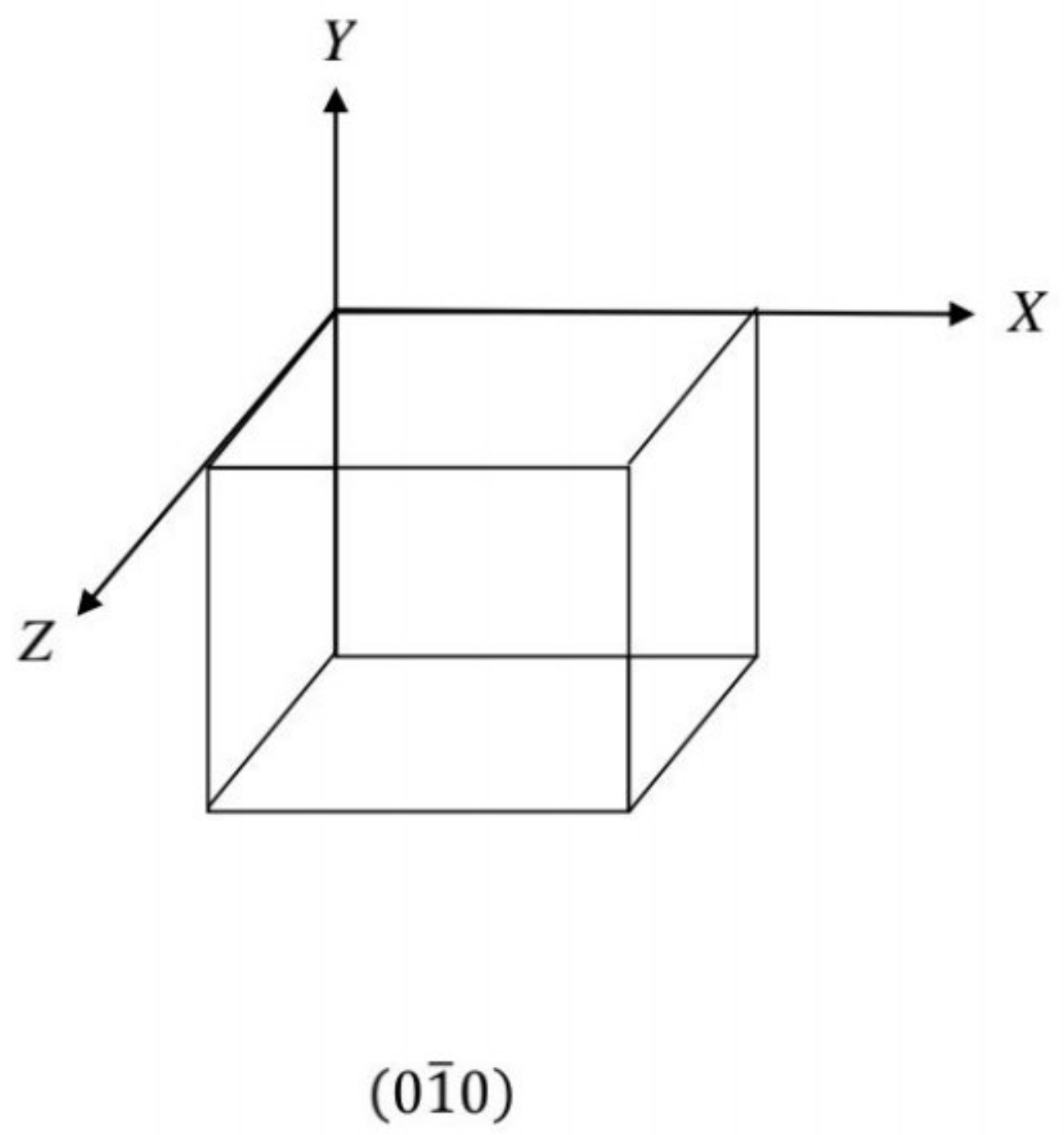
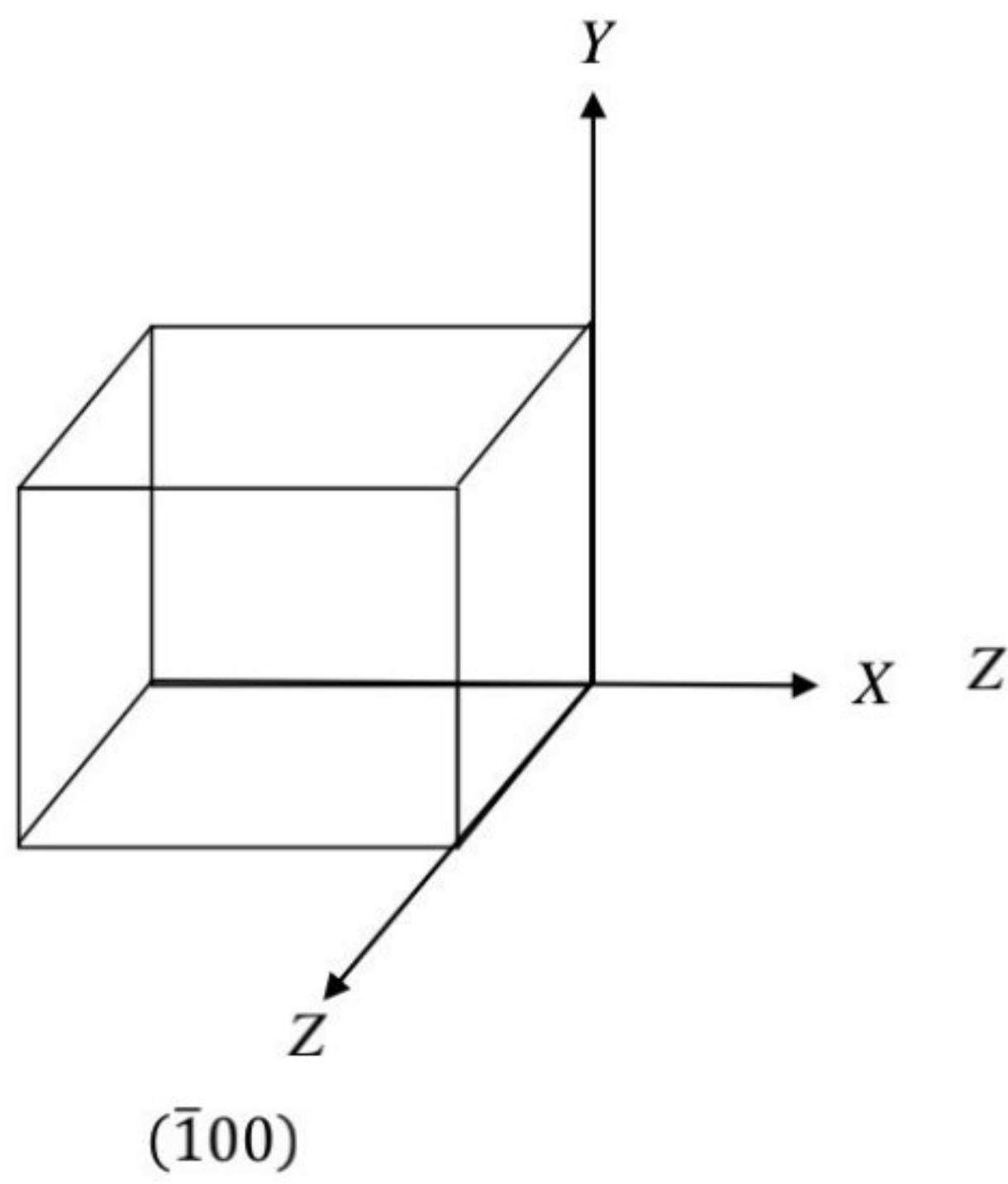
Draw of the Planes: (100) , (010) , (001) , (110) , (011) , (101) , (111) , $(\bar{1}00)$, $(0\bar{1}0)$, $(00\bar{1})$.



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Relation between inter-planar spacing & Miller indices:

Let us consider a plane shown in fig.-03 belongs to a family of planes whose Miller indices are (hkl) .

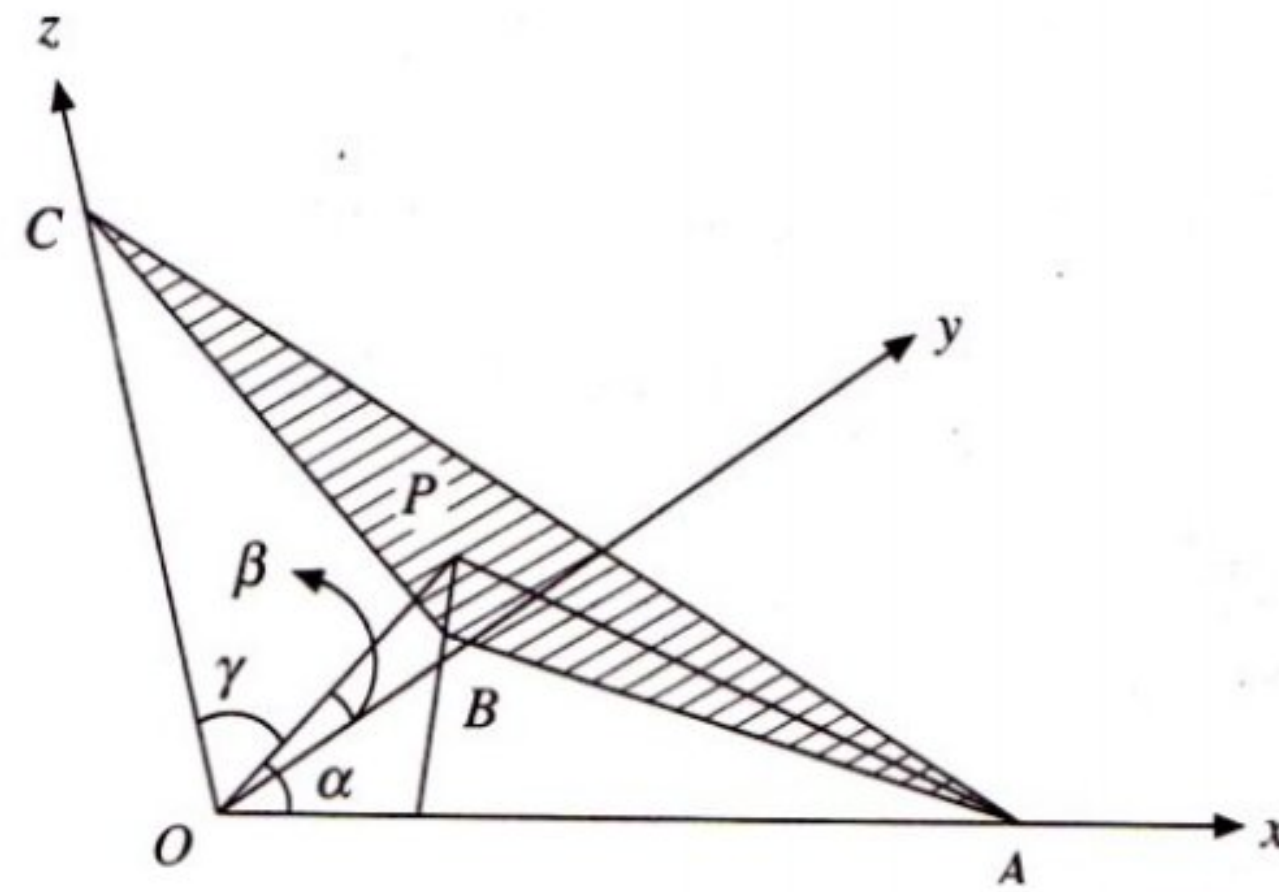


Fig.-03: (hkl) plane intercepting x , y and z axes at A , B and C , respectively.

The perpendicular OP from origin to the plane represents the interplaner spacing d of this family of planes. Let the direction cosines of OP be $\cos\alpha'$, $\cos\beta'$ and $\cos\gamma'$ as shown. The intercepts of the plane on the three axes are $OA = \frac{a}{h}$, $OB = \frac{b}{k}$ and $OC = \frac{c}{l}$, where a, b and c are the length of the cube edge.

Therefore

$$\begin{aligned}\cos\alpha' &= \frac{d}{OA} = \frac{d}{\frac{a}{h}} = \frac{dh}{a} \\ \cos\beta' &= \frac{d}{OB} = \frac{d}{\frac{b}{k}} = \frac{dk}{b} \\ \cos\gamma' &= \frac{d}{OC} = \frac{d}{\frac{c}{l}} = \frac{dl}{c}\end{aligned}$$

where, $d = OP$ represents perpendicular distance between adjacent members of the same family of planes.

According to the rule of cosine direction,

$$\begin{aligned}\cos^2\alpha' + \cos^2\beta' + \cos^2\gamma' &= 1 \\ \left(\frac{dh}{a}\right)^2 + \left(\frac{dk}{b}\right)^2 + \left(\frac{dl}{c}\right)^2 &= 1\end{aligned}$$

For cubic lattice, $a=b=c$.

$$\therefore d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Defects in Solids:

The term Defect denotes a state of the lattice in which the atomic arrangement has departed from regularity. There are different kinds of defects;

- i. Point defects
- ii. Line defects
- iii. Plane defects
- iv. Electronic Imperfections
- v. Excitation states of crystals
- vi. Transient imperfections

The descriptions of defects are as follows:

1. **Point defects:** If the deviation from regularity is confined to a very small region of only about a few lattice constants, it is called a point defect. It is of four types:
 - i) Interstitial atom
 - ii) Schottky defect or vacancy
 - iii) Frenkel defect
 - iv) Substitutional impurity atom
2. **Line defects:** The example of this defect is dislocation. It is a region of strain in the crystal in which, part of the lattice has undergone a shearing strain equal to one lattice vector called a Burgers vector. It is of two types.
3. **Plane defects:** These defects have an extension in area and are confined to a small region. There are three types of plane defects:
 - i) Lineage boundary
 - ii) Grain boundary
 - iii) Stacking boundary
4. **Electronic Imperfections:** Errors in charge distribution or energy in solids are called electronics defects. These defects are responsible for important electrical and magnetic properties.
5. **Excitation states of crystals:** These are imperfections as these cause deviations from perfect crystal symmetry. Some examples are
 - i) Phonons and magnons
 - ii) Conduction electrons and holes
 - iii) Excitons
 - iv) Quantized plasma waves
6. **Transient imperfections:** These imperfections are introduced into the crystal from external sources, i.e.
 - i) Photons
 - ii) High energy charged particles like, electrons, protons, mesons and ions.

High energy unchanged particles such as neutron and neutral

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Crystal binding:

Crystal binding is formed by the attractive electrostatic interaction between the negative charges of the electrons and the positive charges of the nuclei which is entirely responsible for the cohesion of solids. Magnetic forces have only a weak effect on cohesion and gravitational forces are negligible.

The strength of crystal bond is measured by the energy required to dissociate a solid into isolated atoms, molecules or ions as appropriate and the energy is generally known as cohesive energy.

Different types of crystal bindings:

1. **Ionic Bond:** Ionic crystals are made up of positive and negative ions. The ionic bond results from the electrostatic interaction of oppositely charged ions.
Example: NaCl, CsCl, etc.
2. **Co-valent Bond:** The co-valent bond is the classical electron pair or homopolar bond of chemistry, particularly of organic chemistry. The cohesive forces in co-valent crystals arise from the sharing of electrons by adjacent atom. Each atom that participates in a co-valent bond contributes an electron to the bond.
Example: Diamond, Silicon, Germanium, etc.
3. **Metallic Bond:** The valence electrons of metal atoms are weakly bound. When such atoms interact to become a crystal, their valence electrons form a 'gas' of electron that move with relative freedom through the resulting assembly of metal ions. The characteristic feature of the metallic binding is the lowering of the energy of the valence electrons in the metal as compared with the free atom.
Example: Cu, Ag, Au, etc.
4. **Van der waals Bond:** All atoms and molecules exhibit weak, short-range attractions for one another due to van der waals forces. The bond formed by the van der waals interaction is known as van der waals bond. Van der waals interaction are responsible for the condensation of gases into liquids and freezing of liquids into solids in the absence of ionic, co-valent or metallic bonding mechanisms.
Example: He, Ne, Ar, etc.
5. **Hydrogen Bond:** A neutral hydrogen atom has only one electron. It should form a co-valent bond with only one other atom. However, under certain condition an atom of hydrogen is attracted by rather strong forces to two atoms, thus forming a hydrogen bond between them, with a bond energy of the order of 0.1 eV. A hydrogen bond occurs between certain molecules containing hydrogen atoms.
Example: HF_2^- , H_2O etc.

